

Anabel Yong

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Summary

MSc in Computational Statistics & Machine Learning from UCL with expertise in Gaussian Processes, Bayesian Optimisation, and kernel-based methods. Interested in machine learning for life sciences discovery and learning biological data.

Education

MSc Computational Statistics & Machine Learning, UCL (Distinction, Dean's List) Sept 2023 - Sept 2024

Modules: Probabilistic & Unsupervised Learning (Gatsby PhD module) (91%), Machine Learning Seminar (Marc Deisenroth- DeepMind)(76%), Numerical Optimization (80%), Statistical Natural Language Processing

BSc Biochemistry (with Mathematics), University of Edinburgh October 2019 - June 2023

Modules: Bioinformatics (95%), Machine Learning (80%), Numerical Methods for Differential Equations (85%), Linear Programming, Modelling & Solutions (76%), Mathematics for Natural Sciences (80%)

Work Experience

- **Visiting Machine Learning Research Scholar, Jonathan Scarlett** May 2024 - Present
 - Implemented biological sequence kernels, Inverse Quadratic Hamming kernels for amino acid sequences.
 - Designing an active learning algorithm for mutation data.
- **Machine Learning Research Intern, IgnotaLabs.AI, Dr.Layla-Hosseini-Gerami, Dr.Austin Tripp, Cambridge MLG Group & Brooks Paige, UCL** June 2024- Sept 2024
 - Developed a novel multi-objective Bayesian optimization algorithm (GP-MOBO) using Tanimoto/MinMax Kernels for small molecule optimization. (Paper pending submission.)
 - Deployed a Python package (KERN-GP), enabling hardware-optimized Gaussian processes for molecular property prediction, with scalability tested against GPU-based PyTorch models.
- **Research Assistant, Kustatscher Lab, University of Edinburgh x Technical University of Munich** Sept 2022 - July 2023
 - Benchmarked a data pre-processing pipeline for proteomics, pgFDR), to enhance protein detection accuracy.
 - Aided in database curation of human proteins and microproteins of ProteomeHD2
- **Statistical Genetics Intern, ZiHeng Yang Lab, UCL Centre of Computational Biology, UCL** March 2022 - May 2022
 - Developed Bayesian MCMC algorithms to model ABO blood type frequencies, contributing to population genetics and personalised medicine approaches.
- **Research Intern, Grima Lab, Department of Systems & Synthetic Biology, University of Edinburgh** June 2020 - Jan 2022
 - Implemented stochastic and deterministic models to predict mRNA decay kinetics, aiding in the understanding of gene regulation mechanisms and supporting drug efficacy studies in preclinical research.

Personal Projects (on Portfolio)

- **DoLa-based Decoding for Instruction-Following LLMs**
 - Evaluated and adapted DoLa contrastive decoding in T5 and FLAN-T5, enhancing instruction accuracy and reducing hallucinations for large language models (LLMs).

Skills

Programming: Python, C++, R, MATLAB, git, Linux

Machine Learning: PyTorch, JAX, BoTorch, GPyTorch

Bioinformatics/Cheminformatics: RDKit, PyMOL, UniProt, MaxQuant

Cloud Computing & HPC: Google Cloud Platform (GCP), AWServices (AWS)